

ISO26262 Test Case Sheet

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Development Method: **ATDD**

Field	Value
Document Template	test_case_sheet
Development Method	ATDD
Requirement	What is fault injection testing in ISO 26262?
Related ISO Parts	Part 11 - Semiconductors Part 5 - Hardware Development
Referenced Pages	64, 65, 77, 82, 83

ISO26262 Evidence

Part: Part 11 - Semiconductors

Clause: Unknown

Page: 64

Score: 0.7948

Summary: ■ ISO 26262-11:2018(E) ■ type of safety mechanism including required confidence level; ■ observation points and diagnostic points; ■ fault site, fault list; and ■ workload used during fault injection. In particular, the verification planning describes and justifies: ■ the fault model and the related level of abstraction: ■ As clarified in the following clauses for DFA, digital, analogue and PLD, fault injection can be performed at the appropriate level depending on the fault model being considered, the specific semiconductor technology, feasibility, observability and use case; and NOTE 1 Depending on the purpose, fault injection can be implemented at different abstraction levels (e.g. semiconductor component top-level, part or subpart level, RTL, etc.). A rationale for the abstraction level

Part: Part 11 - Semiconductors

Clause: 5.1.10

Page: 82

Score: 0.779

Summary: ■ ISO 26262-11:2018(E) 5.1.10 Verification using fault injection simulation 5.1.10.1 General As mentioned in 4.8, fault injection is a useful method for semiconductor components. This is especially true for digital circuits for which fault insertion testing of single-event upsets at the hardware level is impractical or even impossible for certain fault models. Therefore, fault injection using design models (e.g. fault injection done at the gate-level netlist) is helpful to complete the verification step. NOTE 1 Fault injection can be used both for permanent (e.g. stuck-at faults) and transient (e.g. single-event upset) faults. NOTE 2 Fault injection is just one of the possible methods for verification, and other approaches are possible. Fault injection utilizing design models can be successful

Part: Part 11 - Semiconductors

Clause: Unknown

Page: 65

Score: 0.7724

Summary: ■ ISO 26262-11:2018(E) EXAMPLE 7 When verifying the diagnostic coverage of a dual-core lock-step, the relevant fault population could be limited to the compared CPU outputs and related fault locations. EXAMPLE 8 When verifying the diagnostic coverage of a software-based hardware test, CPU internal faults are relevant. NOTE 6 Techniques like fault collapsing can also be used to reduce the faults population to prime faults. ■ the fault injection controls, with respect to the related claim in the respective safety analysis; and EXAMPLE 9 Fault injection controls can include the type of fault to be injected, the duration of a transient fault, the number of faults injected in a simulation run, time and location of fault occurrence and the window of observation of the expected action of a safety

Part: Part 5 - Hardware Development

Clause: 4.8

Page: 77

Score: 0.7421

Summary: ■ ISO 26262-11:2018(E) NOTE 7 Due to the complexity of modern digital components (millions of gates), fault injection methods can assist the computation and be used for verification of the amount of safe faults and especially of the failure mode coverage. See 4.8 and 5.1.10 for details. Fault injection is not the only method, and other approaches are possible as described in 5.1.10. 5.1.7.2 How to consider transient faults of digital components 5.1.7.2.1 Failure rate of transient fault As described in ISO 26262-5:2018, 8.4.7, NOTE 2, the transient faults are considered when shown to be relevant due, for instance, to the technology used. They can be addressed either by specifying and verifying a dedicated target “single-point fault metric” value to them or by a qualitative rationale. NOTE A

Part: Part 11 - Semiconductors

Clause: 5.1.2

Page: 83

Score: 0.7322

Summary: ■ ISO 26262-11:2018(E) If properly exercised, methods derived from stuck-at simulations (like N-detect testing, see for example References [35] to [37]) can be applied for verification of non-stuck-at fault models as well. EXAMPLE 3 Since exhaustiveness is not required, the non-stuck-at fault models analysis can be applied to a subset of the digital component subparts selected depending on their possible impact (for example comparators) or on a statistical basis. EXAMPLE 4 For N-detect testing, “properly exercised” means that N different detections of the same fault are guaranteed by the pattern set (i.e. pattern richness). N can range from 5 to 10. NOTE Fault injection can also be used to inject bridging faults (see 5.1.2) in specific locations based on layout analysis or to verify the im

Hazard / ASIL

Stuck-at faults, single-event upsets (SEUs), bridging faults

ASIL: ■■ ■■

Safety Goal / FSR / TSR

Safety Goal:

Verification of diagnostic coverage and failure mode coverage for semiconductor components

FSR:

Failure mode coverage and safe fault identification

TSR:

Test bench workload and fault injection parameters

ATDD Acceptance Criteria

TC-001 - Verify dual-core lock-step comparator fault detection

Given: Dual-core lock-step comparator operating under normal conditions

When: Inject stuck-at fault at CPU output

Then: System detects fault and triggers safety mechanism

Expected Action: Safety mechanism activates within 100ms

Verification Method: Fault injection simulation using gate-level netlist

Methodology: ATDD

Traceability

Development Method: ATDD

Requirement: REQ-001

Hazard: HZ-001

Safety Goal: SG-001

FSR: FSR-001

TSR: TSR-001
Test Cases: TC-001

Trace Chain:
REQ-001 → HZ-001 → SG-001 → FSR-001 → TSR-001 → TC-001